

Mauricio Vilar Giribet

Data Science and Software Developer

Python and FastAPI · React and TypeScript · SQL and NoSQL · ETL and data quality · Plotly Dash, Tableau and Power BI

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PROFESSIONAL SUMMARY

Developer who works **both halves of the same job**: I build the software, and I do the analysis that software exists to serve. On the software side I write **Python and FastAPI** services and REST APIs under **React, TypeScript and Next.js** front ends, backed by PostgreSQL, MySQL, MongoDB, Supabase and Redis, and I ship them as Docker containers and serverless functions on Azure, AWS, Render and Vercel with Git based workflows. On the data side I own the reporting cycle end to end: sourcing from databases, REST APIs and spreadsheets, cleaning and validation, **SQL modelling and extract, transform and load (ETL) flows**, descriptive and inferential statistics, and dashboards in **Plotly Dash, Tableau and Power BI**, with **24 published analyses** behind that half. The two are not separate skills for me. My pipelines hold up because I write them as software, with **automated tests and traceability** of where every figure comes from, and my applications are useful because I already know how the person on the other side reads a number. My degree is in Business Administration, which is why I turn a business need into a specification instead of waiting for one. **The job title has kept changing; the work I end up doing has not.** Native Spanish, B2/C1 English.

CORE SKILLS

Software Development Python · FastAPI · TypeScript and JavaScript (ES6+) · React · Next.js · REST API design and versioning · JSON data handling and schema validation · HTML5, CSS3 and Bootstrap · Git and GitHub, branching and code review · automated testing with pytest · VS Code, Cursor, GitHub Copilot and Claude Code

SQL, Databases and Data Engineering SQL (joins, CTEs, aggregations, subqueries and window functions) · PostgreSQL · MySQL · **NoSQL**: MongoDB, Supabase and Redis · data modelling for reporting and for application state · extract, transform and load (ETL) · integration and consolidation of multiple sources · REST APIs and webhooks · automation of recurring loads · Databricks · Parquet and pyarrow

Data Science, Analysis, Statistics and Modelling Python (pandas, NumPy, scikit-learn) · exploratory analysis · descriptive and inferential statistics · hypothesis testing (Mann-Whitney U, Welch) · A/B testing · funnel, cohort and retention analysis · revenue metrics (LTV and ARPU) · anomaly detection with thresholds derived from the data · churn prediction and customer segmentation (logistic regression, random forest, K-means)

Data Quality, Validation and Traceability Quality and completeness checks against specification · cleaning · detection and resolution of inconsistencies, duplicates and missing values · reconciliation across sources · traceability of figures · automated quality tests · procedural documentation

Visualisation and Reporting Plotly Dash · Tableau · Power BI with DAX and data modelling · dashboard design and maintenance · definition and tracking of key performance indicators (KPIs) · advanced Excel (pivot tables, advanced formulas, macros, Power Pivot and financial models) and Power Query · executive reporting and data storytelling for non technical stakeholders

Cloud, Containers and AI Integration Docker · serverless functions · Microsoft Azure (Functions, AI Foundry, Azure OpenAI, Key Vault, Entra ID) · AWS EC2 · Render and Vercel · self hosted n8n · secrets handling and RBAC · large language model integration (Anthropic, OpenAI and Azure OpenAI APIs) · LangChain · Retrieval Augmented Generation (RAG) · Model Context Protocol (MCP) servers

PROFESSIONAL EXPERIENCE

Software Engineer

Jul 2026 to Present

Dot Com Media (DCM Moguls) · Remote, California, USA

- Clean up and restructure the workflows in GoHighLevel and in the company's internal tools, and build and operate production data flows and automations that integrate the CRM, databases and REST APIs, debugging their behaviour and performance against real usage data.
- Develop new features from scratch for the internal software products, and build the entire front end of several of those products end to end.
- Work in English with a distributed team, delivering against deadlines across several concurrent projects.

Founder, Data and Software

Oct 2025 to Present

NyxAI Studio · Mexico City

- Run a data and software consultancy for companies in Mexico, owning the full cycle: gathering needs with the client, scoping, technical specification, build, production deployment and support.
- Develop full stack web applications: React, TypeScript and Next.js front ends over Python and FastAPI services and REST APIs, backed by PostgreSQL, Supabase, MongoDB and Redis.
- Build the data layer those applications run on: integrate CRMs, spreadsheets, databases and REST APIs, clean and validate the information, and turn it into the indicator reports and dashboards business owners use to decide, with each procedure documented before it is automated so every figure stays traceable.
- Deploy and operate the infrastructure: Docker containers, serverless functions and self hosted n8n on Azure and AWS, with identity management, RBAC and secrets handling.

Business Development Associate <i>Cenote Gardens · Tulum, Mexico</i> - Designed and built commercial dashboards from HubSpot CRM data to track key performance indicators (KPIs) for lead generation, conversion and closing, owning the data integrity of the funnel end to end. - Built financial models (IRR, NPV and cash flow projection) to support pricing decisions with investors from Mexico, the United States and Europe, while leading the commercial team.	Jun 2024 to Jul 2025
Independent trader, assets and crypto assets <i>Self employed · Mexico City</i> Develop quantitative analysis and backtesting scripts in Python over time series and live market data, under a formal risk framework (position sizing, stop levels and reward to risk ratio).	Jan 2020 to Present
Professional internships <i>Add Valant and Hotel Four Points by Sheraton · Puebla, Mexico</i> Updated and validated client databases, correcting inconsistencies before operational use, and handled invoicing and collections according to internal procedures and committed deadlines.	Jan 2023 to Dec 2023
SELECTED TECHNICAL PROJECTS	
Credit risk: leakage testing, metric choice and production feature engineering	
Exploratory analysis of 30,000 card holders to predict next month payment default, over an imbalanced target of 22 % defaults, which is why the metrics were chosen up front: ROC AUC, PR AUC, Gini and KS instead of accuracy. Includes an explicit data leakage test that rules out the three routes it could enter by. I then built the production feature pipeline: a scikit-learn transformer that derives 15 variables from six monthly snapshots of balance, payment and delinquency, and a ColumnTransformer that imputes, scales and encodes by cardinality, with every derived variable validated by its univariate ROC AUC before being allowed in.	
Digital product analytics: SQL, A/B testing and cohorts	
End to end analysis combining SQL and Python: conversion funnels, cohort retention and revenue metrics (LTV and ARPU), with A/B tests and hypothesis testing, closing with an executive report of actionable recommendations. The dashboard shipped in Plotly Dash as a web application, usable with no installation by non technical stakeholders.	
AI labour market: source auditing and measurement with confidence intervals	
- Pipeline of my own in Python that collects job postings from the Adzuna API, classifies them with a custom taxonomy and normalises salaries with the World Bank purchasing power factor. Medians by metro area with 90 % bootstrap confidence intervals and a minimum sample size rule of 30 observations, over 3,781 postings, covered by 85 automated tests. - Discarded a public 50,000 row dataset after detecting from its shape that it was generated, and filtered out the salaries the provider models rather than publishes.	
Statistical modelling, segmentation and churn, TripleTen	
Analysed prepaid telecom usage logs in Python and SQL to identify churn drivers, compute ARPU and recharge frequency, and propose segmented retention actions. Segmented platform users with K-means on standardised data, using silhouette analysis to set the optimal number of clusters.	
EDUCATION	
Data Science and Data Analytics Bootcamp <i>TripleTen</i>	2025
B.A. in Business Administration <i>Universidad de las Americas Puebla (UDLAP), Puebla, Mexico</i>	2024
CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT	
A2 · Excel and macros, Power Pivot and DAX, and Power BI	2025
DataCamp · Introduction to FastAPI and Introduction to APIs in Python	2026
DataCamp · Introduction to Databricks	2026
DataCamp · Retrieval Augmented Generation (RAG) with LangChain and Developing LLM Applications with LangChain	2026
DataCamp · Working with the OpenAI API and Developing AI Systems with the OpenAI API	2026
DataCamp · DevOps Concepts and Software Development with Claude Code	2026
Microsoft Learn · Azure AI: Document Intelligence, Azure OpenAI and Azure AI Foundry	2026
Anthropic · Official Claude courses: Claude Code, Prompt Engineering, Building with Claude, MCP and Tool Use	2025
TripleTen · Data Science Bootcamp achievements: Dashboards Whiz (Tableau and Dash) and Business Analysis Guru (business metrics and A/B testing)	2025
DataCamp · Power BI: Data Preparation, Data Transformation and Customer Churn Case Study	2022
LANGUAGES	
Spanish: native. English: B2/C1; natively bilingual through Mexican and Australian dual nationality.	